

Alice Perego

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Ph.D. Student in Astrophysics
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MAIN SCIENTIFIC INTERESTS

LISA mission - ultra-compact double white dwarfs - multi-messenger astrophysics - photometric and spectroscopic surveys - LISA data analysis - stellar binary evolution - Milky Way - gravitational wave backgrounds - massive black hole binaries - extreme mass-ratio inspirals

EDUCATION

Observatoire de la Côte d'Azur – Artemis

Nice, France

Ph.D. in Astrophysics

November 2024 – present

- Thesis title: *Future perspectives on gravitational-wave astrophysics with space-based detectors: enabling multi-messenger studies of Galactic binaries*
- Supervisors: Nelson Christensen, Astrid Lamberts
- Expected Graduation: Autumn 2027

University of Milano Bicocca

Milan, Italy

Master's Degree in Astrophysics and Space Physics

October 2022 – October 2024

- Final grade : 110/110 cum laude
- Thesis title: *Assessing the performance of future space-based detectors: astrophysical foregrounds and individual sources*
- Supervisors: Alberto Sesana, Matteo Bonetti, Sylvain Marsat

Laboratoire des 2 infinis (L2IT)

Toulouse, France

Research internship

March 2024 – July 2024

- Project title: *Performing parameter estimation of massive black holes binaries for future space-based gravitational wave detectors, in the perspective of the project Voyage2050.*
- Advisor: Sylvain Marsat

University of Milano Bicocca

Milan, Italy

Bachelor's Degree in Physics

October 2019 – September 2022

- Final grade : 110/110 cum laude
- Thesis title: *Numerical simulations of oscillating neutron stars*
- Supervisor: Bruno Giacomazzo

CONFERENCES, WORKSHOP & RESEARCH VISITS

- European Astronomical Society Annual Meeting
Contributed talk
- Lausanne, Switzerland – July 2026

- LISA Symposium College Park, USA – June 2026
Contributed talk
- LISA France Paris, France – May 2026
Contributed talk
- LISA Astrophysics Working Group meeting Geneva, Switzerland – December 2024
Contributed talk
- GW Space 2050 workshop Potsdam, Germany – November 2025
Oral presentation
- “Enabling Future Gravitational Wave Astrophysics
in the Milli-Hertz Regime” workshop Garching, Germany – July 2025
Oral presentation
- LISA Sprint workshop Huntsville, USA – April 2025
- LISA Distributed Data Processing Center (DDPC) workshop Milan, Italy – June 2025
- Research visit at Carnegie Mellon University Pittsburgh, USA – April 2025
Host: Katelyn Breivik
- Annual meeting of the French Society of Astronomy & Astrophysics Toulouse, France – July 2025
Contributed talk – published proceeding: [Preparing for LISA with CFH telescope: Synergies between gravitational waves and electromagnetic observations](#)
- Rencontres de Moriond - Gravitation La Thuile, Italy – April 2025
Contributed talk – published proceeding: [Assessing the performance of future space-based detectors: astrophysical foregrounds and individual sources](#)
- LISA Astrophysics Working Group meeting Garching, Germany – November 2024
Contributed talk
- GWOSC Gravitational Wave Open Data workshop Toulouse, France – April 2024
Mentor at an in-person study hub

SEASONAL SCHOOLS & SPECIALISED TRAINING

- LISA School for early-career scientists Les Houches, France – October 2025
- MOOC “Reproducible Research II:
Practices and tools for managing computations and data” Online – September 2026
- MaNiTou Summer School on Gravitational Waves Toulouse, France – July 2024
- Masterclass on Big Data within Science and Industry Milan, Italy – September 2023

MEMBERSHIP

- LISA Consortium - core member
- LISA DDPC - L3C coordination unit
- LISA Astrophysics working group
- GW Space 2050 working group

OBSERVATIONAL PROPOSALS

- ‘Apapane: Dynamic Sky Across All Scales
PIs: Viraja Khatu, Alice Perego, Damya Souami submitted in May 2026
Canada France Hawaii Telescope/Megacam – [300-700] nights
- Spectroscopic follow-up of double white dwarf candidates
PI: Alice Perego submitted in September 2026
ESO/UVES – 152 hours

SOFTWARE SKILLS

- **Programming:** Python, C++
- **Softwares and codes:** Standard scientific Python packages (astrophysics, data analysis, machine learning), COSMIC, LISABETA, GBGPU, EXTRAPOPS, (population synthesis and GW data analysis), Vizier and Aladin (astronomical dataset and visualisation tools)
- **Analysis techniques:** Frequentist and Bayesian statistical inference

LANGUAGES

- **Italian:** Native
- **English:** Fluent
 - Certification: Bbetween English C1 Open Badge awarded by the University of Milano Bicocca
- **French:** Intermediate
 - Certification: French proficiency test B2 awarded by the GoFluent online course

PUBLICATIONS AND RESEARCH OUTPUTS

Perego, A., Bonetti, M., Sesana, A., Toonen, S. & Korol, V. Assessing the performance of future space-based detectors: Astrophysical foregrounds and individual sources. *Physical Review D* **114**, 023022. doi:[10.1103/nvgj-8pmc](https://doi.org/10.1103/nvgj-8pmc). arXiv: [2510.18695](https://arxiv.org/abs/2510.18695) [[astro-ph.HE](#)] (July 2026).

Perego, A., Lamberts, A., Schultheis, M. & Christensen, N. Towards systematic searches for LISA white dwarf binaries with multiband photometry. *Astronomy & Astrophysics* **701**, L6. doi:[10.1051/0004-6361/202556494](https://doi.org/10.1051/0004-6361/202556494). arXiv: [2509.09348](https://arxiv.org/abs/2509.09348) [[astro-ph.HE](#)] (Sept. 2025).

Perego, A. et al. Crossmatching LISA detections of white dwarf binaries with electromagnetic catalogues. *in prep.*

Blas, D. *et al.* GUEST: Gravitational Universe Exploration with Satellite Tracking. A passive satellite laser-ranging mission for the dark gravitational Universe. *arXiv e-prints*, arXiv:2607.18390. doi:[10.48550/arXiv.2607.18390](https://doi.org/10.48550/arXiv.2607.18390). arXiv: [2607.18390](https://arxiv.org/abs/2607.18390) [[astro-ph.CO](#)] (July 2026).

Coughlin, S. *et al.* COSMIC-PopSynth/COSMIC: v3.6.1 version v3.6.1. July 2025. doi:[10.5281/zenodo.15855958](https://doi.org/10.5281/zenodo.15855958).